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Results, Visualization, and Demonstration

APS

NOV 2026
CHECKPOINT

ADAPTIVE PATHOGEN SURVEILLANCE | INDIVIDUAL PROGRESS BRIEF

MY ROLE

I focus on explaining what our simulation does and what its results mean. My part is to turn the facility, sensor readings, source predictions, and evaluation numbers into understandable visuals and a clear project demonstration.

OUR WORKING FOUNDATION

Our current pipeline can show the true source of a simulated event, the baseline prediction, and the error between them. I am organizing how to present that information clearly. Later visuals will compare machine-learning and adaptive methods when those experiment results are available.

A single run makes a good demonstration, but repeated trials are needed before I can describe overall model performance.

MY CHECKPOINT STATUS

I am: [building / testing / refining]

MY CONTRIBUTION SO FAR

I have completed: [specific file, test, or output]

TERMS AND IDEAS I USE

A facility network diagram draws zones as points and the connections between them as lines. Ground truth is the actual simulated source, while the predicted source is the model's answer. Exact-source accuracy is the share of runs predicted perfectly. Graph-distance error measures how many connections away an incorrect prediction is. Static sampling uses the original readings; adaptive sampling adds one selected reading before updating the prediction.

WHAT I AM WORKING ON NEXT

1. I will draft a facility diagram showing zones, connections, available readings, the true source, and the predicted source.
2. I will build clear tables or charts for accuracy and graph-distance error, then compare methods under noise and missing-data conditions.
3. I will outline a short demonstration that moves from simulated event to readings, prediction, and evaluation. A simple Streamlit screen is optional.

CHALLENGES I AM CONSIDERING

My visuals depend on consistent outputs from the simulation, data, and model components. I need to label results accurately and avoid presenting an example as if it proves overall performance.

WHAT I PLAN TO SHOW

I plan to bring a draft facility illustration or chart, plus a short sequence for explaining the final demonstration.

HOW I WOULD EXPLAIN MY WORK

My checkpoint summary: My contribution is making APS understandable to someone who has not seen the code. I want to show where the simulated event started, what the sensors observed, what the system predicted, and how close that prediction was. As model results arrive, I will help the team present fair comparisons.